



A First Look At Google's Go Programming Language

Go is a new systems programming language launched by Google, and has received wide attention in the programming community. This article gives you an overview of this language, with some specific examples to help understand its features.

Go was announced just a few months back, but is already being touted as the next C language. Why?

The 'C' language itself evolved from a language known as 'B'. C was created in the 1970s, and still continues to be widely used; however, the language has mostly stopped evolving, and there is a dire need for a new language that could replace C. There are many languages that have been named 'D' (the most popular being the one by Walter Bright), or those that want to be the 'D' language. Still, nothing has made the cut, so far. Go might well become the next C language—or it may be 'Gone' in a few years!

Is there substance behind the hype and buzz around Go? Yes, a lot! Most systems programmers (like me) find it very good after trying it out and writing non-trivial programs. Why?

- Go has a familiar C-like syntax (but not exactly C syntax, as we'll see later)
- Yet, Go has the capabilities of modern dynamic languages like Python.
- Go is statically typed and almost as efficient as C (the performance of Go programs is within

10-20 per cent of the equivalent C code)!

- Though the language is designed for systems programming, it has capabilities like garbage collection and reflection, which make it a powerful language. Go is not an object-oriented language, but it has (arguably) novel features like interfaces, as we'll see later in this article.

Go has already won Tiobe's 'Language of the Year Award 2009'. Tiobe (<http://www.tiobe.com/>) is one of the most widely-referred-to programming language popularity indexes.

There is a lot to cover on Go, so I'll be limiting myself to the most important aspects. First I'll cover the essential background information, and then I'll present some sample programs to introduce language features.

What is Go?

Go is a new, experimental, concurrent, garbage-collected, systems programming language.

- **New and experimental:** It is a new and evolving language that is still at the experimental stage. No production systems have yet been developed using Go.